

DIRICHLET STEREOHEDRA: CERTIFIED LOWER BOUNDS AND LOCAL RIGIDITY

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ABSTRACT. The largest known number of facets of a convex polyhedron that tiles $\mathbb{R}^3$ by congruent copies is 38, attained by Engel’s Dirichlet stereohedra for the space group $I4_132$ (1981). We make four contributions.

- (1) We point out that the statement “38 is the maximum for plesiohedra”, which appears in standard references, is not proved in the cited source. That source is a numerical grid search.
- (2) We search all space groups with numbers 75–230 again, including lattice metrics outside the previously sampled range, and certify every facet count in exact rational arithmetic. We find no Dirichlet stereohedron with more than 38 facets.
- (3) For ten space groups we obtain new certified lower bounds for the maximal number of facets of their Dirichlet stereohedra. For example, $I4_1/a$ has a Dirichlet stereohedron with 29 facets; the best previously reported value was 23.
- (4) We prove, by an exact computation, that Engel’s 38-facet tile is locally rigid. Near it there is no isohedral tile with the same combinatorics that is not a Dirichlet cell. Single non-regular flips of the dual complex admit no convex realization nearby.

1. THE PROBLEM AND ITS STATUS

Problem (G) of Grünbaum and Shephard [4] asks for the least upper bound on the number of facets of a convex polytope that tiles $\mathbb{R}^n$ by congruent copies. For $n \geq 3$ it is open, and it is not even known whether such a bound exists.

For a stereohedron, that is, the tile of an isohedral tiling, Delone’s theorem bounds the number of facets by $2^n(a + 1) - 2$, where a is the number of aspects. In $\mathbb{R}^3$ this gives 390, and Tarasov improved it to 378. A *Dirichlet (Voronoi) stereohedron* is the Voronoi cell of one point x in its orbit Γx under a space group Γ . For these, Bochiş, Santos and Sabariego [1, 2, 5, 6] proved bounds group by group. The largest of their bounds is 92, for the quarter cubic groups.

The record of 38 facets is due to Engel [3]. Schmitt [7] computed the Dirichlet stereohedra of all 145 tetragonal, trigonal, hexagonal and cubic space groups on fine point grids. He confirmed Engel’s four combinatorial types with f -vector $(70, 106, 38)$ and found nothing larger. Wikipedia and MathWorld cite [7] for the claim that 38 is the maximum for plesiohedra. However, §2.3 of [7] describes a grid search, and its §2.1 states problem (G) as open. As far as we can tell, **the maximality of 38 is not proved.**

Two features of the grid search matter below.

- The 38-facet chamber of $I4_132$ is covered by only 153 of about 10^9 grid points, so chambers smaller than a grid cell could be missed.
- For non-cubic groups the ratio c/a was sampled only in $[\frac{1}{2}, \frac{7}{2}]$, with about 10^6 position points for each value.

2. EXACT CERTIFICATES

We work in fractional coordinates with the metric tensor $M = B^\top B$. This tensor is rational whenever $(c/a)^2$ is rational, also for hexagonal lattices. The Voronoi cell of a rational point x is

$$D(x) = \{z : 2(y-x)^\top Mz \leq y^\top My - x^\top Mx \text{ for all } y \in \Gamma x \setminus \{x\}\}.$$

A *certificate* that $D(x)$ has at least k facets consists of k distinct bisector planes, each containing three non-collinear points of $D(x)$ that are verified exactly to be feasible. The points are vertices found in floating point and then recomputed exactly as intersections of three planes. It suffices to use the orbit points y with $|y-x| < 2\rho_L$, where ρ_L is the covering radius of the lattice. Indeed, $D(x)$ is contained in the Voronoi cell of the lattice $x+L \subset \Gamma x$.

The method reproduces Engel's (70, 106, 38) at Schmitt's point (427, 761, 1421)/6984, and Schmitt's value 35 for $I4_122$ at $c/a = 727/500$, both exactly.

3. SEARCH AND NEW LOWER BOUNDS

For each of the 156 groups with numbers 75–230 we proceeded in three stages.

- (1) We sampled 8000 random points, with $c/a \in [0.3, 4]$ for non-cubic groups.
- (2) We ran 25 local hill-climbs, which maximise the facet count and, as a tie-breaker, minimise the gap to the nearest non-facet bisector. This steers the search into small chambers.
- (3) We certified the best points exactly.

For the 71 groups without mirror planes we also searched $c/a \in [0.05, 0.5]$ and $c/a \in [3.5, 20]$.

Floating-point runs reported up to 58 facets. These were artefacts: points within about 10^{-12} of a mirror plane, where nearly coincident planes were counted twice. The exact certificates at the same places give Schmitt's values. We report only certified numbers.

Proposition 1. *The Dirichlet stereohedra at the rational points listed in Table 1 have at least the stated numbers of facets. In each of these ten groups this exceeds the maximum found in [7].*

IT	group	[7]	certified	point x (fractional)	c/a
86	$P4_2/n$	20	21	(52/961861, 121892/661793, 83579/121807)	10891/5403
88	$I4_1/a$	23	29	(1, 923317/1000000, 5661/250000)	5847/9646
110	$I4_1cd$	22	24	(67153/89224, 250857/935476, 160855/736201)	89519/250000
122	$I42d$	31	33	(499999/999999, 143779/371659, 492461/895745)	8925/6004
138	$P4_2/ncm$	14	17	(454227/1000000, 954227/1000000, 758507/1000000)	4162/9787
167	$R\bar{3}c$	26	27	(460309/760898, 209157/996041, 531553/944363)	21055/9436
184	$P6cc$	11	13	(741399/1000000, 2653/8000, 429439/500000)	3/10
192	$P6/mcc$	11	13	(118179/200000, 180021/250000, 1/2)	2501/7628
201	$Pn\bar{3}$	16	17	(954723/963917, 107169/832985, 156955/308036)	1
228	$Fd\bar{3}c$	16	17	(443121/863615, 155697/999223, 713295/929221)	1

TABLE 1. New certified lower bounds. The certificates are in `certificates.json`, and `verify_certificates.py` rechecks them independently. For IT 138, 184 and 192 the value of c/a lies outside the range $[\frac{1}{2}, \frac{7}{2}]$ sampled in [7].

No certified count exceeded 38, in any group or metric range, and the maximum of 38 was again attained only in $I4_132$. Squashed lattices do not help: the facet maxima occur at moderate $c/a \approx 0.1$ – 0.45 , and as $c/a \rightarrow 0$ the counts decrease.

4. LOCAL RIGIDITY OF DIRICHLET STEREOHEDRA

Fix a tile P of an isohedral face-to-face tiling with group Γ , and fix its combinatorial type. The tiling conditions are:

- **(L)** for each neighbour γ , the map γ takes the vertices of the facet $F_{\gamma^{-1}}$ to those of F_γ ; these equations are linear in the vertices;
- **(P)** every facet is planar.

Every solution near a tiling is again a tiling. Neighbouring copies meet along facets, and the angle sums around edges are continuous multiples of 2π , so they remain equal to 2π . Both conditions are affine-invariant and rational in fractional coordinates. We compute the rank of the Jacobian of (L)+(P) exactly, modulo two 31-bit primes, at exact vertex coordinates. We also compute exactly the tangent directions of the Dirichlet family: moving x , and changing c/a for non-cubic groups.

Theorem 2 (computer-assisted). *For Engel’s 38-facet tile ($I4_132$, $x = (427, 761, 1421)/6984$) the solution set of (L)+(P) has tangent space of dimension exactly 3, spanned by the motions of x . Hence, to first order, every isohedral face-to-face tile near it with the same combinatorics is a Dirichlet stereohedron. The same holds for Schmitt’s 35-facet tile of $I4_122$, with dimension 4 (motions of x and the change of c/a), and for 46 further generic Dirichlet stereohedra in 13 groups with 12 to 38 facets.*

Floating-point versions of this computation suggested one or two extra directions. That was an artefact of edges of length about 10^{-9} .

We also built the Γ -invariant dual complex of Engel’s tiling from the exact cell. A dual cell must contain all orbit points equidistant from the vertex, not only the facet neighbours; the dual cells contain 4 or 6 sites. We then tried all 26 equivariant $2 \rightarrow 3$ flips that give the cell a new neighbour. None admits a nearby realization satisfying (L)+(P) with the new facet opened, even to area 10^{-6} . The residual stays between 10^{-6} and 10^{-2} , and the resulting tiles are not convex. Non-Dirichlet stereohedra with more facets, if they exist, are therefore not obtained by local modification of the Dirichlet records.

CODE

All code and data are at <https://github.com/VladimirMoor/GEOTILES>, in the directory `research/p4`: `dvfast.py`, `exact.py`, `certificates.json`, `verify_certificates.py`, `deform_exact.py`, `rigidity_scan.py` and `flip.py`.

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